

Object Oriented Data Analysis

Short Course: Based on Marron and Dryden (2021)

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Emphasis: object definition → geometry → statistical method.

Roadmap

- 1 OODA foundations
- 2 Curves as data objects
- 3 Distance-based OODA
- 4 Shape analysis and Procrustes methods
- 5 Manifold-valued data and PGA
- 6 Putting it together

The OODA Viewpoint

Progression of statistical objects:

Area	Data Object	Geometry
Univariate statistics	Numbers	$\mathbb{R}$
Multivariate analysis	Vectors	$\mathbb{R}^p$
Functional data analysis	Curves/functions	Function spaces
OODA	General objects	Metric/manifold spaces

Examples of OODA object spaces:

- Shape spaces
- Wasserstein spaces
- Spaces of trees and graphs
- Spaces of covariance matrices
- Spheres and manifolds

Two-hour plan

- | | |
|--|--------|
| 1. Motivation and OODA vocabulary | 15 min |
| 2. Curves as objects: FPCA and registration | 25 min |
| 3. Metric-space methods: means, MDS | 20 min |
| 4. Shape analysis: Procrustes, quotients, Kendall geometry | 30 min |
| 5. Manifold tools: tangent spaces and PGA | 20 min |
| 6. Short exercises and discussion | 10 min |

Learning objectives

By the end, students should be able to:

- identify data objects and object spaces;
- choose a geometry or distance;
- derive PCA/MDS/Procrustes basics;
- explain Fréchet means and tangent-space analysis;
- recognize when Euclidean methods are misleading.

From Big Data to Complex Data

The usual data matrix

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nd} \end{bmatrix}$$

Rows are cases; columns are features. This is powerful, but not neutral.

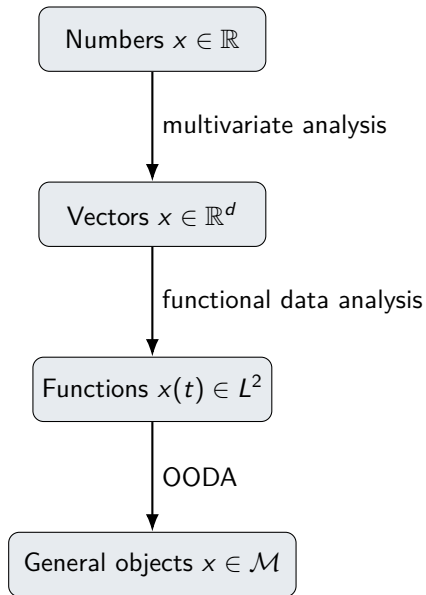
OODA question

What if the observation is not naturally a row vector?

- a curve,
- a shape,
- a tree,
- a graph,
- a covariance matrix,
- a probability distribution.

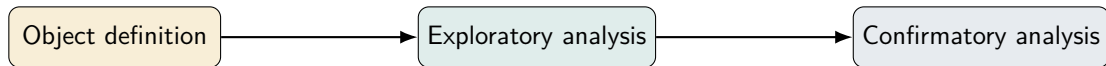
The atomic unit is the **data object**, not necessarily a vector.

The OODA hierarchy



The four OODA questions

1. **Data objects:** What is one observational unit?
2. **Object space:** Where do possible objects live conceptually?
3. **Feature space:** What numerical representation is used computationally?
4. **Methods:** Which distances, means, visualizations, and inferential tools are appropriate?



Pedagogical warning

The hard part is often not the algorithm; it is defining the object in a way that respects the science.

Object space versus feature space

Object space $\mathcal{M}$

The mathematical or conceptual population of possible data objects.

$$X_1, \dots, X_n \in \mathcal{M}.$$

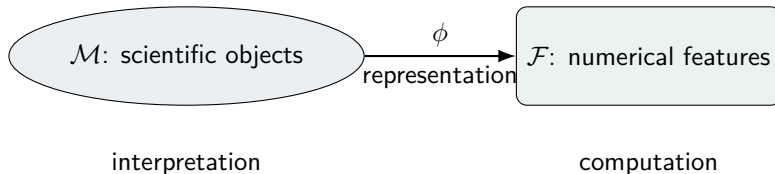
Examples: curves, shapes, trees, SPD matrices.

Feature space $\mathcal{F}$

The numerical representation used for computation.

$$\phi : \mathcal{M} \longrightarrow \mathcal{F}.$$

Examples: sampled grids, landmarks, adjacency matrices, eigenvalues.



Modes of variation

Definition 1 (Mode of variation)

A mode of variation is a one-dimensional, or approximately one-dimensional, set of candidate objects in the object space that summarizes an interpretable component of sample variation.

Linear example

In Euclidean PCA, a mode is

$$\mu + tv_k, \quad t \in \mathbb{R},$$

where v_k is a loading direction.

Nonlinear example

On a manifold, a mode is often a geodesic

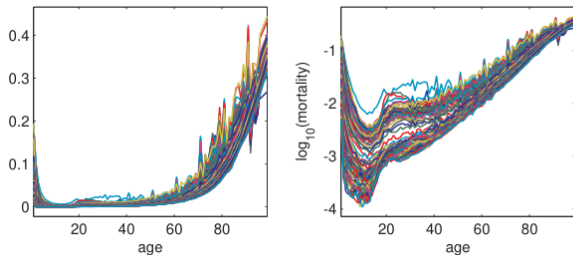
$$\gamma_k(t) = \text{Exp}_\mu(tv_k).$$

In OODA, the mode should be interpretable as a set of objects, not just as coefficients.

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Case study: Spanish mortality curves



Object definition

Mortality table: years $\times$ ages. Two possible object choices:

- age-indexed curves, one per year;
- year-indexed curves, one per age.

Chosen object

$$X_i(t) = \log_{10}\{\text{mortality at age } t \text{ in year } i\}.$$

The log scale reveals structure hidden in raw probabilities.

Functional data model

Let

$$X_1, \dots, X_n \in L^2(\mathcal{T}), \quad \langle f, g \rangle = \int_{\mathcal{T}} f(t)g(t) dt.$$

Mean function:

$$\hat{\mu}(t) = \frac{1}{n} \sum_{i=1}^n X_i(t).$$

Centered data:

$$Y_i(t) = X_i(t) - \hat{\mu}(t).$$

Covariance kernel:

$$\hat{C}(s, t) = \frac{1}{n} \sum_{i=1}^n Y_i(s)Y_i(t).$$

Functional PCA eigenproblem

$$\int_{\mathcal{T}} \hat{C}(s, t)\phi_k(t) dt = \hat{\lambda}_k\phi_k(s), \quad \|\phi_k\|_{L^2} = 1.$$

Derivation: FPCA as variance maximization

Proposition 1 (First functional principal component)

The first loading function solves

$$\phi_1 = \arg \max_{\|\phi\|=1} \frac{1}{n} \sum_{i=1}^n \langle Y_i, \phi \rangle^2.$$

It is the eigenfunction of $\hat{C}$ with largest eigenvalue.

Sketch.

Write

$$V(\phi) = \frac{1}{n} \sum_i \langle Y_i, \phi \rangle^2 = \langle \hat{C}\phi, \phi \rangle.$$

Maximize $V(\phi)$ subject to $\langle \phi, \phi \rangle = 1$. The Lagrangian

$$\mathcal{L}(\phi, \lambda) = \langle \hat{C}\phi, \phi \rangle - \lambda(\langle \phi, \phi \rangle - 1)$$

has first-order condition $\hat{C}\phi = \lambda\phi$.



Scores, loadings, and reconstruction

Scores:

$$\xi_{ik} = \langle Y_i, \phi_k \rangle.$$

Rank- K reconstruction:

$$X_i(t) \approx \hat{\mu}(t) + \sum_{k=1}^K \xi_{ik} \phi_k(t).$$

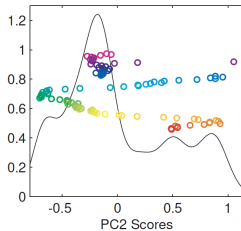
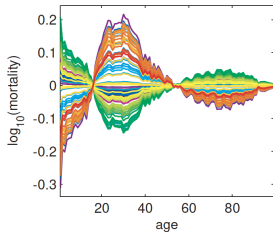
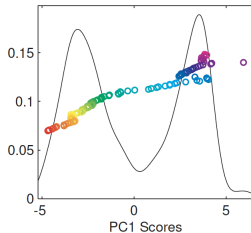
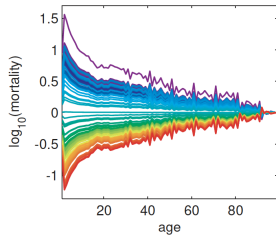
Explained variance fraction:

$$\frac{\sum_{k=1}^K \hat{\lambda}_k}{\sum_{j \geq 1} \hat{\lambda}_j}.$$

Interpretation

- $\phi_k(t)$ tells where on the domain the variation occurs.
- ξ_{ik} tells where object i lies along that mode.
- The pair $(\phi_k, \{\xi_{ik}\})$ gives an interpretable mode of variation.

Mortality PCA: history in two dimensions



PC1

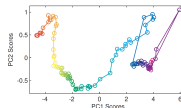
Dominant overall mortality improvement; outliers and reversals are historically interpretable.

PC2

Contrast between young adult males and the rest of the population.

Lesson

Dimension reduction is not only compression.



Amplitude and phase: why ordinary FPCA can fail

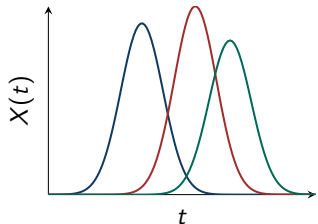
Suppose peaks are shifted in time:

$$X_i(t) = f_i(\gamma_i(t)).$$

Here:

- f_i is the aligned amplitude object;
- γ_i is the phase or warping object.

$$\Gamma = \{\gamma : [0, 1] \rightarrow [0, 1] : \gamma(0) = 0, \gamma(1) = 1, \dot{\gamma} > 0\}.$$



Problem

Misalignment can make PCA spend components explaining horizontal shifts rather than biologically meaningful shape.

Fisher–Rao registration

For an absolutely continuous function f , define the square-root slope function

$$q(t) = \text{sign}\{\dot{f}(t)\} \sqrt{|\dot{f}(t)|}.$$

Warping action:

$$(q, \gamma)(t) = q(\gamma(t)) \sqrt{\dot{\gamma}(t)}.$$

Proposition 2 (Key invariance)

For $q_1, q_2 \in L^2$ and any $\gamma \in \Gamma$,

$$\|(q_1, \gamma) - (q_2, \gamma)\|_{L^2} = \|q_1 - q_2\|_{L^2}.$$

One-line proof.

Use change of variable $u = \gamma(t)$:

$$\int_0^1 |q_1(\gamma(t)) - q_2(\gamma(t))|^2 \dot{\gamma}(t) dt = \int_0^1 |q_1(u) - q_2(u)|^2 du.$$



Registration objective and example

Amplitude distance:

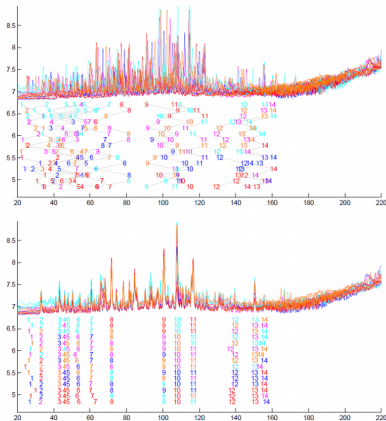
$$d_A(f_1, f_2) = \inf_{\gamma \in \Gamma} \|q_1 - (q_2, \gamma)\|_{L^2}.$$

Estimated alignment:

$$\hat{\gamma} = \arg \min_{\gamma \in \Gamma} \|q_1 - (q_2, \gamma)\|^2.$$

OODA interpretation

The amplitude and phase components are different possible data objects.



TIC mass-spectrometry curves: registration aligns chemically corresponding peaks.

Exercise 1: functional objects

Five-minute discussion

For wearable-sensor heart-rate recordings from n individuals over a day, propose two different data object definitions.

For each proposal, specify:

1. object space $\mathcal{M}$;
2. feature representation $\phi(X)$;
3. natural distance $d(X_i, X_j)$;
4. one exploratory plot;
5. one scientific question it can answer.

Hint

Possible object choices include daily curves, registered activity episodes, distributions of heart-rate values, and subject-specific covariance summaries.

Mini-Exercise 2: Pan Cancer data in R

The dataset are from Hoadley et al. (2014, Cell). There are $p=12478$ genes and $n=300$ individuals, with 50 in each of six groups. The groups are Bladder Cancer (BLCA), Kidney Renal Cancer (KIRC), Ovarian Cancer (OV), Head and Neck Squamous Cell Cancer (HNSC), Colon Adenocarcinoma (COAD), and Breast Cancer (BRCA).

```
x<-read.csv("PanCancer-ReducedGeneExpressionData.csv",header=TRUE)
dim(x)
w<-x[2:12479,2:301] # remove first row and column
w<-sapply(w,as.numeric)
group<-unname(x[1,2:301])
group<-c(as.matrix(group))
table(group)
gcol<-rep(1:6,each=50) # give each group a different color
X<-t(w)
dim(X)
pca <- prcomp(X)
plot(pca$sdev**2/sum(pca$sdev**2))
```

The scree plot indicates that perhaps 10 genes summarize most of the information. Note that in total the number of PCs with positive variability is $\min(n-1, p)$.

Mini-Exercise (continued)

What do PC1 and PC2 represent?

```
plot(pca$x[,1],pca$x[,2],col=gcol,xlab="PC1",ylab="PC2",type="n")
text(pca$x[,1],pca$x[,2],group,col=gcol,cex=0.5)
plot(pca$rotation[,1])
plot(pca$rotation[,2])
plot(apply(X,2,sd),ylab="sd")
```

Carry out PCA with correlation matrix R

```
pca2 <- prcomp(X,scale=TRUE)
plot(pca2$sdev**2/sum(pca2$sdev**2),ylab="percent")
plot(pca2$x[,1],pca2$x[,2],col=gcol,xlab="PC1",ylab="PC2")
plot(pca2$x[,1],pca2$x[,2],col=gcol,xlab="PC1",ylab="PC2",type="n")
text(pca2$x[,1],pca2$x[,2],group,col=gcol,cex=0.5)
```

Which is more appropriate, PCA with S or with R?

Mini-Exercise (continued)

What do PC1 and PC2 represent?

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Carry out PCA with correlation matrix R

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text(pca2$x[,1],pca2$x[,2],group,col=gcol,cex=0.5)
```

Which is more appropriate, PCA with S or with R?

- I think that PCA with S is more appropriate, as all variables are of the same type (gene expression measurement).

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When only distances are available

Many object spaces are most naturally described by a distance:

$$(\mathcal{M}, d), \quad X_1, \dots, X_n \in \mathcal{M}.$$

Examples:

Object	Natural distance
Curves	L^2 , Fisher–Rao
Landmark shapes	Procrustes
Distributions	Wasserstein
Covariance matrices	affine-invariant Riemannian
Graphs/trees	edit, spectral, topological distances

Core questions

What is the center? What is variation? How do we visualize a distance matrix?

Fréchet means

Definition 2 (Sample Fréchet mean)

For observations $X_1, \dots, X_n \in (\mathcal{M}, d)$, define

$$F(p) = \frac{1}{n} \sum_{i=1}^n d^2(X_i, p).$$

Any minimizer

$$\hat{\mu}_F \in \arg \min_{p \in \mathcal{M}} F(p)$$

is a sample Fréchet mean.

Proposition 3 (Euclidean special case)

If $\mathcal{M} = \mathbb{R}^d$ and $d(x, p) = \|x - p\|_2$, then

$$\hat{\mu}_F = \bar{X} = \frac{1}{n} \sum_i X_i.$$

Derivation: Euclidean Fréchet mean

$$F(p) = \frac{1}{n} \sum_{i=1}^n \|X_i - p\|^2 = \frac{1}{n} \sum_i (X_i - p)^T (X_i - p).$$

Differentiate:

$$\nabla_p F(p) = \frac{2}{n} \sum_i (p - X_i) = 2(p - \bar{X}).$$

Thus $\nabla_p F(p) = 0$ implies

$$p = \bar{X}.$$

Hessian:

$$\nabla_p^2 F(p) = 2I_d > 0,$$

so this is the unique minimizer.

But in general metric spaces

The mean may not be unique, may not have a closed form, and may lie in a geometrically surprising location.

Classical multidimensional scaling (MDS)

Given only pairwise distances d_{ij} , construct coordinates $y_i \in \mathbb{R}^r$.

$$D^{(2)} = (d_{ij}^2), \quad H = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T.$$

Double-centered Gram matrix:

$$B = -\frac{1}{2} H D^{(2)} H.$$

Eigen-decomposition:

$$B = U \Lambda U^T.$$

Coordinates using positive eigenvalues:

$$Y_r = U_r \Lambda_r^{1/2}.$$

Interpretation

MDS turns an object-distance matrix into a Euclidean display, approximately preserving geometry.

Derivation: why double-centering works

Assume unknown centered Euclidean coordinates $y_i \in \mathbb{R}^r$ with

$$\sum_i y_i = 0, \quad B_{ij} = y_i^T y_j.$$

Then

$$d_{ij}^2 = \|y_i - y_j\|^2 = B_{ii} + B_{jj} - 2B_{ij}.$$

Matrix form:

$$D^{(2)} = b\mathbf{1}^T + \mathbf{1}b^T - 2B, \quad b = (B_{11}, \dots, B_{nn})^T.$$

Multiplying by H on both sides removes the additive row/column terms:

$$HD^{(2)}H = -2HBH = -2B.$$

Thus

$$B = -\frac{1}{2}HD^{(2)}H.$$

Wasserstein distance for one-dimensional distributions

For probability distributions P, Q on $\mathbb{R}$ with quantile functions F_P^{-1} and F_Q^{-1} ,

$$W_2^2(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|^2 du.$$

Object space

$\mathcal{M} = \{\text{probability distributions on } \mathbb{R}\}.$

Feature space

Use quantile functions:

$$\phi(P) = F_P^{-1} \in L^2[0, 1].$$

Takeaway

For distributions, the distance can encode transport geometry rather than pointwise density differences.

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Shapes as data objects

A landmark configuration is

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_k^T \end{bmatrix} \in \mathbb{R}^{k \times m}, \quad x_j \in \mathbb{R}^m.$$

Shape removes translation, rotation, and scale.

Definition 3 (Shape equivalence)

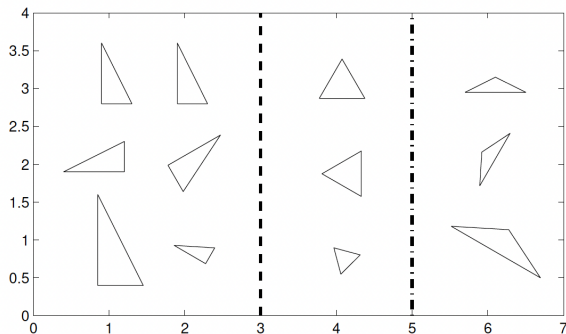
Two configurations $X, Y \in \mathbb{R}^{k \times m}$ have the same shape if

$$Y = cXR + \mathbf{1}a^T$$

for some $c > 0$, $R \in \text{SO}(m)$, and $a \in \mathbb{R}^m$.

$$[X] = \{cXR + \mathbf{1}a^T : c > 0, R \in \text{SO}(m), a \in \mathbb{R}^m\}.$$

Triangle example: quotient geometry



Equivalence classes

Each vertical band shows triangles with the same shape, differing by translation, rotation, and scale.

Quotient space

$$\text{Shape space} = \frac{\text{configurations}}{\text{translation, rotation, scale}}.$$

For planar triangles, this space is a sphere S^2 .

Pre-shape construction

Let $X \in \mathbb{R}^{k \times m}$. Define the centering matrix

$$C = I_k - \frac{1}{k} \mathbf{1} \mathbf{1}^T.$$

Remove translation:

$$X_c = CX.$$

Remove scale:

$$Z = \frac{X_c}{\|X_c\|_F}.$$

The pre-shape space is

$$\mathcal{S}_m^k = \{Z \in \mathbb{R}^{k \times m} : \mathbf{1}^T Z = 0, \|Z\|_F = 1\}.$$

Dimension count

Translation removes m dimensions; scaling removes 1:

$$\dim(\mathcal{S}_m^k) = km - m - 1.$$

Shape space further quotients rotations, usually removing $m(m-1)/2$ dimensions.

Procrustes alignment problem

Given centered, scaled configurations $Z_1, Z_2 \in \mathcal{S}_m^k$, align rotations by

$$\min_{R \in \text{SO}(m)} \|Z_1 - Z_2 R\|_F^2.$$

Expand:

$$\|Z_1 - Z_2 R\|_F^2 = \|Z_1\|_F^2 + \|Z_2\|_F^2 - 2 \text{tr}(Z_1^T Z_2 R).$$

Since $\|Z_1\|_F = \|Z_2\|_F = 1$, this is equivalent to

$$\max_{R \in \text{SO}(m)} \text{tr}(AR), \quad A = Z_1^T Z_2.$$

Proposition 4 (Orthogonal Procrustes)

If $A = U\Sigma V^T$, then an optimal rotation is approximately

$$\hat{R} = VU^T,$$

with the usual determinant correction when $\det(VU^T) < 0$.

Derivation: Procrustes via SVD

Let $A = U\Sigma V^T$. For $R \in SO(m)$,

$$\text{tr}(AR) = \text{tr}(U\Sigma V^T R) = \text{tr}(\Sigma V^T RU).$$

Set

$$Q = V^T RU.$$

Then Q is orthogonal, so $|q_{jj}| \leq 1$, and

$$\text{tr}(AR) = \sum_{j=1}^m \sigma_j q_{jj} \leq \sum_{j=1}^m \sigma_j.$$

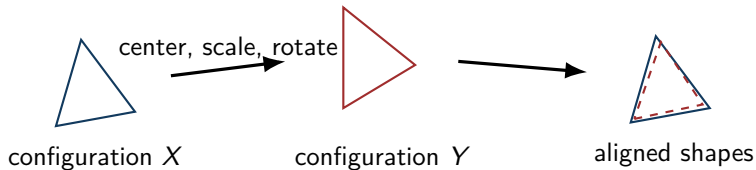
The upper bound is achieved when $Q = I$, hence

$$V^T RU = I \implies R = VU^T.$$

Geometry

Procrustes distance is not just a formula; it is the distance after optimizing over nuisance transformations.

Visualizing Procrustes alignment



Statistical consequence

After alignment, residual variation estimates true shape variation rather than location, orientation, or size variation.

Generalized Procrustes Analysis (GPA)

Algorithm 1 (GPA)

Given configurations $X_1, \dots, X_n$:

1. Center and scale each configuration to pre-shape Z_i .
2. Initialize a template $\mu^{(0)}$.
3. For iteration $r = 0, 1, 2, \dots$:
 - 3.1 align each Z_i to $\mu^{(r)}$ by Procrustes rotation;
 - 3.2 average the aligned configurations;
 - 3.3 center and scale the average to obtain $\mu^{(r+1)}$.
4. Stop when the template changes negligibly.

What GPA estimates

A mean shape and aligned residuals for PCA, clustering, or inference.

Kendall shape-space geometry

After pre-shape normalization, rotations still act on the right:

$$Z \mapsto ZR, \quad R \in SO(m).$$

Kendall shape space is the quotient

$$\Sigma_m^k = \mathcal{S}_m^k / SO(m).$$

Planar shapes

For k landmarks in $m = 2$ dimensions:

$$\Sigma_2^k \cong \mathbb{C}P^{k-2}.$$

For $k = 3$ landmarks:

$$\Sigma_2^3 \cong \mathbb{C}P^1 \cong S^2.$$

Key idea

Even simple shape data lead naturally to curved non-Euclidean spaces.

Exercise 2: Procrustes by hand

For two centered and scaled planar configurations $Z_1, Z_2 \in \mathbb{R}^{k \times 2}$, suppose

$$A = Z_1^T Z_2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}.$$

1. Write the Procrustes problem as a trace maximization.
2. Compute the SVD of A numerically or symbolically.
3. Find the optimal rotation $\hat{R}$.
4. Interpret what happens if $\det(\hat{R}) = -1$.

Tutorial extension

Repeat with Z_2 reflected. Does allowing reflections change the fitted distance?

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Manifold-valued objects

A Riemannian manifold $\mathcal{M}$ is locally Euclidean but globally curved. Examples:

- directions: S^{p-1} ;
- planar triangle shapes: S^2 ;
- SPD matrices: $\mathcal{S}_{++}^p$;
- skeletal models: $\mathbb{R}^{3k} \times \mathbb{R}_+^\ell \times (S^2)^m$.

Geodesic distance:

$$d(p, q) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\|_{\gamma(t)} dt.$$

Geodesic equation:

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

Exponential and logarithm maps

Exponential map

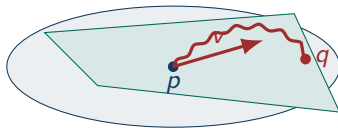
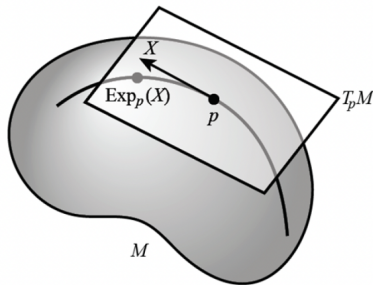
$$\text{Exp}_p : T_p\mathcal{M} \rightarrow \mathcal{M}.$$

Given tangent vector v , follow the geodesic from p with initial velocity v for unit time.

Logarithm map

$$\text{Log}_p(q) \in T_p\mathcal{M}.$$

Returns the tangent vector pointing from p to q .



$T_p\mathcal{M}$ approximates $\mathcal{M}$ near p

Intrinsic mean and first-order condition

$$\hat{\mu} = \arg \min_{p \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n d^2(p, X_i).$$

If all points lie away from the cut locus and the mean is unique, the gradient condition is

$$\sum_{i=1}^n \text{Log}_{\hat{\mu}}(X_i) = 0 \in T_{\hat{\mu}}\mathcal{M}.$$

Why this generalizes the Euclidean mean

In Euclidean space,

$$\text{Log}_p(X_i) = X_i - p,$$

so

$$\sum_i (X_i - p) = 0 \iff p = \bar{X}.$$

Principal Geodesic Analysis (PGA)

Algorithm 2 (Tangent-space PGA)

1. *Compute an intrinsic or extrinsic mean $\hat{\mu} \in \mathcal{M}$.*
2. *Map observations to the tangent space:*

$$v_i = \text{Log}_{\hat{\mu}}(X_i) \in T_{\hat{\mu}}\mathcal{M}.$$

3. *Perform ordinary PCA on $v_1, \dots, v_n$.*
4. *Map principal directions back:*

$$\gamma_k(t) = \text{Exp}_{\hat{\mu}}(te_k),$$

where e_k is a tangent-space PC direction.

Approximation

Tangent-space PCA is local. It works best when data are concentrated relative to curvature.

Medical imaging example: skeletal objects

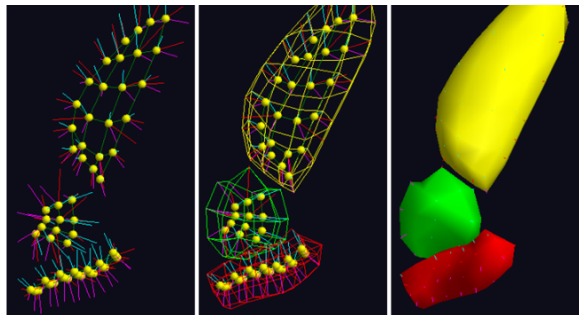


Figure 1.10: Skeletal representation of a single bladder-prostate-rectum

Bladder-prostate-rectum example

Objects are skeletal anatomical representations used for radiation-treatment planning.

Parameters include:

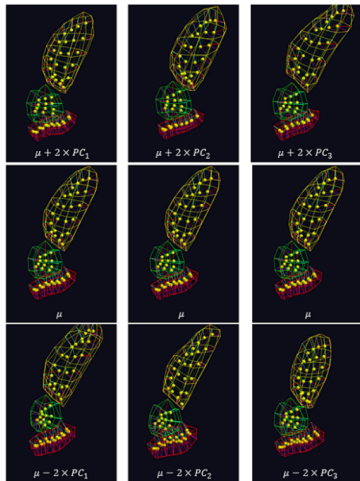
atom locations $\in \mathbb{R}^{3k}$,

spoke lengths $\in \mathbb{R}_+^\ell$, spoke directions $\in (S^2)^m$

Thus the object space is approximately

$$\mathbb{R}^{3k} \times \mathbb{R}_+^\ell \times (S^2)^m.$$

PGA modes in the anatomical example



Interpretation

Columns show three PGA modes:

- vertical shift driven by rectum position;
- twisting of the organ complex;
- bladder filling/emptying.

OODA lesson

A mode is a set of plausible anatomical objects, not just an eigenvector in a parameter table.

SPD covariance matrices as objects

Let

$$\mathcal{S}_{++}^p = \{A \in \mathbb{R}^{p \times p} : A = A^T, A > 0\}.$$

A common Riemannian distance is

$$d_R(A, B) = \left\| \log(A^{-1/2} B A^{-1/2}) \right\|_F.$$

Euclidean issue

The straight-line average may be sensitive to the chosen coordinates and does not respect affine transformations.

Geometric issue

Eigenvalues represent axes of variability; positivity and curvature matter.

Applications: diffusion tensors, brain connectivity, finance, covariance descriptors in computer vision.

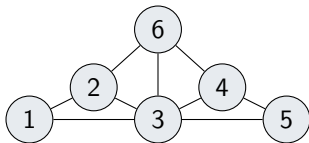
Graph-structured data objects

For a graph $G = (V, E)$, common feature representations include:

- adjacency matrix A ;
- degree matrix D ;
- graph Laplacian $L = D - A$;
- normalized Laplacian $D^{-1/2}LD^{-1/2}$;
- spectra, embeddings, persistence summaries.

Spectral idea

Eigenvectors of L reveal connectivity, bottlenecks, and community structure.



Roadmap

- 1 OODA foundations
- 2 Curves as data objects
- 3 Distance-based OODA
- 4 Shape analysis and Procrustes methods
- 5 Manifold-valued data and PGA
- 6 Putting it together

A general OODA recipe

1. **Scientific goal:** What is the inferential target?
2. **Object definition:** What is one observation?
3. **Geometry:** What transformations are nuisance? What distance respects them?
4. **Center:** Mean, median, template, prototype, or Fréchet center?
5. **Variation:** PCA, PGA, MDS, clustering, regression, classification?
6. **Validation:** resampling, permutation, external replication, sensitivity to object definition.

Key habit

Always ask: which conclusions are properties of the object space, and which are artifacts of the feature representation?

Mini-case design exercise

Choose one of the following contexts:

1. Korean hospital CT/MRI organ-segmentation study;
2. wearable sensors from athletes;
3. topic networks from Korean-language corpora;
4. single-cell gene-expression distributions;
5. covariance matrices from fMRI connectivity.

For the chosen context, specify:

- data object;
- object space;
- feature space;
- distance or metric;
- exploratory method;
- confirmatory question.

Common pitfalls

Pitfall 1: vectorizing too early

Flattening images, shapes, or graphs can hide invariances and distort distances.

Pitfall 2: confusing representation with object

A shape is not its landmark matrix; a distribution is not a histogram bin count; a graph is not only an adjacency matrix.

Pitfall 3: over-interpreting exploratory structure

Low-dimensional displays are useful, but require validation.

Good practice

Run sensitivity analyses over reasonable object definitions and distances.

Take-home messages

1. OODA begins by defining the atomic unit of analysis: the data object.
2. Geometry is not decoration; it determines means, distances, modes, and tests.
3. PCA generalizes in multiple directions: FPCA, MDS, PGA, spectral methods.
4. Shape data naturally lead to quotient spaces and Procrustes methods.
5. Manifold methods are often local, so curvature and concentration matter.
6. Good OODA alternates between scientific interpretation and mathematical structure.

Recommended follow-up reading



Dryden, I. L. and Mardia, K. V. (2016). *Statistical Shape Analysis, with Applications in R*. Wiley.



Marron, J. S. and Dryden, I. L. (2022). *Object Oriented Data Analysis*. Chapman & Hall/CRC.



Ramsay, J. O. and Silverman, B. W. (2005). *Functional Data Analysis*. Springer.



Srivastava, A. and Klassen, E. (2016). *Functional and Shape Data Analysis*. Springer.



Fletcher, P. T., Lu, C., Pizer, S. M. and Joshi, S. (2004). Principal geodesic analysis for the study of nonlinear statistics of shape.

Roadmap

- 7 Appendix: additional derivations and demos

Appendix A: PCA in matrix form

Let $X \in \mathbb{R}^{n \times d}$ be centered. The sample covariance matrix is

$$S = \frac{1}{n} X^T X.$$

PCA solves

$$\max_{\|v\|=1} v^T S v.$$

Lagrangian:

$$\mathcal{L}(v, \lambda) = v^T S v - \lambda(v^T v - 1).$$

First-order condition:

$$S v = \lambda v.$$

Scores:

$$z = X v.$$

Rank- K approximation:

$$X \approx Z_K V_K^T.$$

Appendix B: Procrustes distance variants

Full Procrustes distance:

$$d_F([X], [Y]) = \min_{\beta > 0, R, a} \|X - \beta YR - \mathbf{1}a^T\|_F.$$

Partial Procrustes distance on pre-shapes:

$$d_P([Z_1], [Z_2]) = \min_{R \in SO(m)} \|Z_1 - Z_2 R\|_F.$$

Riemannian shape distance:

$$d_R([Z_1], [Z_2]) = \arccos \left(\max_{R \in SO(m)} \text{tr}(Z_1^T Z_2 R) \right).$$

Practical note

Different distances can produce different means and different PCA/PGA directions.

Appendix C: Central limit intuition on manifolds

If $\hat{\mu}_n$ is an intrinsic mean and μ is the population mean, then under regularity conditions

$$\sqrt{n} \operatorname{Log}_{\mu}(\hat{\mu}_n) \Rightarrow N(0, \Sigma_{\mathcal{M}})$$

in the tangent space $T_{\mu}\mathcal{M}$.

Interpretation

Inference is often carried out after mapping local perturbations of the mean into a tangent space.

Caution

Curvature, nonuniqueness of means, and cut loci can break simple Euclidean intuition.

Appendix D: R demonstration scaffold

```
# Functional PCA workflow sketch
# X: n x d matrix of curves sampled on a common grid
Xc <- scale(X, center = TRUE, scale = FALSE)
sv <- svd(Xc / sqrt(nrow(Xc)))
loadings <- sv$v
scores <- Xc %*% loadings
varfrac <- sv$d^2 / sum(sv$d^2)

# Plot first mode of variation
mu <- colMeans(X)
phi1 <- loadings[,1]
s <- sd(scores[,1])
matplot(grid, cbind(mu - 2*s*phi1, mu, mu + 2*s*phi1),
         type = "l", lty = 1, xlab = "t", ylab = "function")
```

Appendix E: Procrustes demonstration scaffold

```
center_scale <- function(X) {  
  X <- sweep(X, 2, colMeans(X))  
  X / sqrt(sum(X^2))  
}  
  
procrustes_rotate <- function(Z1, Z2) {  
  A <- t(Z1) %*% Z2  
  s <- svd(A)  
  R <- s$v %*% t(s$u)  
  if (det(R) < 0) {  
    s$v[, ncol(s$v)] <- -s$v[, ncol(s$v)]  
    R <- s$v %*% t(s$u)  
  }  
  R  
}  
  
Z1 <- center_scale(X1)  
Z2 <- center_scale(X2)  
Rhat <- procrustes_rotate(Z1, Z2)  
Z2_aligned <- Z2 %*% Rhat
```

Appendix F: possible tutorial sheet

1. Prove the MDS double-centering formula.
2. Derive the Procrustes SVD solution and implement it.
3. Simulate shifted Gaussian peaks and compare PCA before and after registration.
4. Compute Fréchet means on S^1 for concentrated and antipodal samples.
5. Compare Euclidean and affine-invariant distances for SPD matrices.
6. Design an OODA analysis plan for a graph data set.

Questions?

Thank you.